Individual elicitation:

Library Director — individual list

- The system shall report the number of books currently checked out.
- The system shall report which books are most popular.
- The system shall report the number of active members.
- The system shall support tracking late returns so the library can assess whether they decrease.
- The system shall be affordable for a small library to operate and maintain without on-site IT staff.
- The system shall be designed so that e-books or inter-library loans could be added in the future.
- The system shall support a late-return policy that is configurable once the library decides what that policy will be.

Librarian — individual list

- The system shall allow staff to search for books by title or author.
- The system shall allow staff to search for members by name.
- The system shall display whether a book is available or checked out using up-to-date lending information.
- The system shall allow staff to check books in and out in a small number of steps.
- The system shall alert staff when a member has an overdue book.
- The system shall calculate any overdue consequence according to the library's approved policy.

Library Member — individual list

- The system shall allow members to check whether a book is available remotely.
- The system shall allow members to view the books they currently have borrowed and their due dates.
- The system shall allow members to place a hold on a book that is checked out.

- The system shall allow a member to renew a borrowed book when no other member is waiting for it.
- The system shall let members access their account using their library card number without requiring a complicated new account.
- The system shall send members a reminder before a borrowed book is due.
- The system shall support a grace period or other non-fine approach if that is the policy approved by the library.

Record these as unresolved elicitation points

- **Late-return policy conflict:** The Librarian wants daily fines; the Member opposes fines and prefers reminders or a grace period. The Director has not decided. Do not treat either fines or a grace period as settled.
- **Reminder details:** The role card asks for reminders but does not specify when they should be sent or by what method.
- **Hold rules:** The role card says members can hold checked-out books, but does not specify queue order, pickup deadlines, or notifications.
- **Renewal rules:** Renewal is allowed when nobody else is waiting, but the number of renewals and any other eligibility rules are unspecified.
- **Access and privacy:** Card-number access is requested, but the role card does not say how identity would be verified or what account information members can see.
- **Cost and success measures:** The Director wants an affordable system and evidence of fewer late returns, but no budget limit or measurement method is given.

Group Consolidation requirements list

Consolidated requirements list

1. **Reporting: The system shall provide the Director with reports on current checkouts, book popularity, and active member counts.**
2. **Affordability and maintenance: The system shall be affordable and maintainable without on-site IT staff. The budget and maintenance expectations need to be defined.**
3. **Future capabilities: The system shall be designed to support possible future additions such as e-books and inter-library loans.**
4. **Staff search: The system shall allow staff to search for books by title or author and members by name.**
5. **Book availability: The system shall show staff whether a book is available or checked out, with status reflecting recorded loans and returns.**
6. **Checkout and return workflow: The system shall support staff in recording book checkouts and returns through an efficient, straightforward workflow.**
7. **Overdue handling: The system shall identify overdue books and support the library's overdue policy. The policy is unresolved: the Librarian favors fines, while the Member favors reminders and a grace period.**
8. **Member account access: The system shall allow members to view their borrowed books and due dates using their library card number. Security and privacy details need to be clarified.**
9. **Remote access: The system shall allow members to check book availability, place holds, and request renewals remotely. Renewals are allowed only when no other member has placed a hold.**
10. **Due-date reminders: The system shall send members reminders before books are due. Reminder timing and delivery method need to be decided.**
11. **Overdue-policy configuration: The system shall allow an authorized administrator to configure the overdue policy once the library decides which policy parameters apply.**

#	Requirements	Source	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes

1	**Generate reports on current checkouts, book popularity, and active member counts.**	**Library Director**	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**“Popularity” and “active member” need definitions; report formats, filters, and date ranges are unspecified.**
2	**Be low-cost and maintainable without on-site IT staff.**	**Library Director**	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**“Low-cost” and “maintainable” are subjective. Define budget and maintenance expectations.**
3	**Support future addition of e-books and inter-library loans.**	**Library Director**	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**“Support” is vague; the required architectural qualities and integration expectations are unspecified.**
4	**Let staff search for**	**Librarian**	**Pass**	**Fail**	**Fail**	**Pass**	**Pass**	**Pass**	**Search behavior**

	books by title/author and members by name.								**is underspecified (for example, partial matches, sorting, and duplicate names), but basic search can be tested.**
5	**Display current book availability (on-shelf vs. checked-out).**	**Librarian**	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**"Current" and how the system handles reserved, lost, or otherwise unavailable books need definition.**
6	**Facilitate check-in/out in a minimal number of steps.**	**Librarian**	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**"Minimal" is subjective; specify a maximum number of steps or transaction-time target.**
7	**Alert staff to overdue books**	**Librarian / Library Memb**	Fail	**Fail**	**Fail**	**Fail**	**Fail**	**Pass**	**The overdue consequences are**

	and process consequences according to the undecided policy.	**er (conflicting views)**							**unresolved; “alert” lacks timing, delivery, and handling rules.**
8	**Let members view borrowed books and due dates via library card number, without a new, complex account.**	**Library Member**	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**“Complex” is subjective; card number alone may not establish secure identity. Define authentication and account-access requirements.**
9	**Let members remotely check availability, place holds, and renew books if no other holds exist.**	Library Member	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**Hold queues, renewal limits, availability updates, and what “remotely” covers need definition.**

10	**Send automatic reminders before books are due.**	**Library Member**	**Pass**	**Fail**	**Fail**	**Pass**	**Fail**	**Pass**	**Reminder timing, delivery method, retry behavior, and member contact preferences are unspecified.**
11	**Let administrators configure the overdue/ grace-period policy**	**Library Member**	**Pass**	**Fail**	**Fail**	**Fail**	**Fail**	**Pass**	**The policy is unresolved, and it is unclear which parameters can be configured or who is authorized to change them.**

Requirement 7 — Overdue handling

Conflict, completeness, and testability failed. The Librarian favors fines to encourage timely returns, while the Member prefers reminders and a grace period; neither position is selected as the final policy. This remains an open decision for the Library Director or designated decision-maker, who should define the policy and its parameters before the requirement can be made complete and testable.

Requirement 8 — Member account access

Ambiguity, completeness, and testability failed. Replace "using their library card number" as a complete access specification with a requirement to let members securely view their loans and due dates after authentication; the accepted authentication method and privacy controls remain open for the Director to approve. That makes the intended function clearer without assuming that a card number alone is sufficient authentication.

Requirement 10 — Due-date reminders

Ambiguity, completeness, and testability failed. Clarify the requirement to specify that reminders are sent before a loan's due date but leave the exact timing and delivery method (such as email or SMS) as open decisions for the Director. Once those choices are made, the reminder behavior can be stated in measurable, testable terms.

Requirement 11 — Overdue-policy configuration

Completeness failed. Keep the requirement but identify the policy settings the system must support only after the Director resolves the overdue-policy conflict—for example, whether the library uses fines, a grace period, or both. Until the decision is made, this requirement remains provisional rather than implying that configurability itself resolves the stakeholders' disagreement.

AI Comparison

AI-generated requirements list

1. The system shall store and manage records for the library's books.
2. The system shall store and manage records for library members.
3. The system shall record book loans and returns.
4. The system shall allow authorized users to find book and member records.
5. The system shall show whether a book is available or checked out.

6. The system shall maintain accurate, up-to-date records of the library's collection and loans.
7. The system shall provide a way to back up and restore library data.
8. The system shall be simple to use and cost-effective for the library.
9. The system shall support approximately 3,000 books and 800 members.

#	Requirement	Source	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes
1	Store and manage records for the library's books.	AI	Pass	Fail	Fail	Pass	Fail	Pass	"Manage" is broad; required book details and permitted operations are unspecified.
2	Store and manage records for library members.	AI	Pass	Fail	Fail	Pass	Fail	Pass	"Manage" is broad; required member details and privacy/access rules are unspecified.
3	Record book loans and returns.	AI	Pass	Fail	Fail	Pass	Pass	Pass	Basic loan and return recording can be tested, but

									transaction details and rules are missing.
4	Allow authorized users to find book and member records.	AI	Pass	Fail	Fail	Pass	Fail	Pass	“Authorized users” and search behavior are undefined. The scenario does not specify user roles or access permissions.
5	Show whether a book is available or checked out.	AI	Pass	Fail	Fail	Pass	Pass	Pass	The basic status can be tested, but the brief does not define how other statuses or timing are handled.
6	Maintain accurate, up-to-date records of the collection and loans.	AI	Pass	Fail	Fail	Pass	Fail	Pass	“Accurate” and “up-to-date” lack measurable criteria.

7	Provide a way to back up and restore library data.	AI	Pass	Fail	Fail	Pass	Fail	Pass	A plausible operational requirement, but it is an assumption rather than a stated scenario need. Backup frequency and recovery expectations are undefined.
8	Be simple to use and cost-effective for the library.	AI	Pass	Fail	Fail	Pass	Fail	Pass	“Simple” and “cost-effective” are subjective; no usability or budget criteria are given.
9	Support approximately 3,000 books and 800 members.	AI	Pass	Pass	Fail	Pass	Pass	Pass	The stated scale is clear, but the brief does not specify expected growth, performance, or

									whether these numbers are minimums or current totals.

The AI-generated list focuses on digitizing the library's existing records and basic circulation processes. The stakeholder-based list adds more specific needs—such as reporting, remote member services, reminders, future capabilities, and the overdue-policy disagreement—that are not included in the scenario-only list.

7. Comparison Write-u

The consolidated requirements list reflected needs that emerged from role-playing the Library Director, Librarian, and Library Member. The AI-generated list, based only on the scenario brief, covered the basics of digitizing the library's records: managing book and member information, recording loans and returns, finding records, and showing book availability. The lists therefore overlap on core library operations, but the role-play surfaced important needs and disagreements that were not apparent from the brief alone.

Most notably, the AI list missed the conflict over overdue books. The Librarian favored consequences such as fines to encourage timely returns, while the Member preferred reminders and a grace period rather than punitive charges. The Director's role also raised the need for a policy decision that the system could support or configure. The AI's general requirement to record loans and returns did not capture this disagreement or the unresolved policy details. Role-play also surfaced needs absent from the AI list, including reports for the Director, efficient front-desk workflows, remote access for members, holds and renewals, and due-date reminders.

The AI included a backup-and-restore requirement, even though the scenario brief and stakeholder role cards did not explicitly establish it as a need. It is a plausible operational consideration, but in this exercise, it should be identified as an assumption to validate rather than treated as a confirmed requirement. The AI also used broad terms such as "simple to use," "cost-effective," and "accurate" without defining measurable criteria. Some of these ideas connect to stakeholder concerns—for example, the Director wanted an affordable, maintainable system—but the AI's phrasing does not specify what would satisfy those needs.

The lists agree on the need to digitize book and member records, track loans and returns, and make book availability known. The AI also usefully made the system's approximate scale explicit: about 3,000 books and 800 members. That capacity consideration was not stated as a separate requirement in the consolidated list, so it could be worth validating and adding if the library expects the system to support at least that current collection and membership. Overall, the AI list provided a useful baseline, while stakeholder role-play revealed more specific user needs, operational concerns, and a policy conflict requiring further discussion.

8. Final Requirements List

1. **Book records:** The system shall allow authorized library staff to create, view, update, and remove book records.
2. **Member records:** The system shall allow authorized library staff to create, view, update, and remove member records.
3. **Search:** The system shall allow staff to search for books by title or author and members by name.
4. **Availability:** The system shall display whether each book is available or checked out, using current loan and return records.
5. **Loans and returns:** The system shall allow staff to record book checkouts and returns and update the book's availability accordingly.
6. **Efficient staff workflow:** The system shall support staff in completing common checkout and return transactions with a straightforward workflow.
7. **Member account information:** The system shall allow members to view their borrowed books and due dates using their library card number. Authentication and privacy details require further definition.
8. **Remote services:** The system shall allow members to check book availability, place holds, and request renewals remotely. A renewal shall be permitted only when no other member has placed a hold on the book.
9. **Due-date reminders:** The system shall send members reminders before borrowed books are due. The delivery method and timing require further definition.
10. **Overdue handling:** The system shall identify overdue books and support the library's approved overdue policy. The policy—including any fines, grace period, and reminder rules—must be decided with the relevant stakeholders before implementation.

11. **Reports:** The system shall generate reports on current checkouts, book popularity, and active member counts.
12. **Policy configuration:** The system shall allow an authorized administrator to configure overdue-policy parameters, once the library has decided which parameters apply.
13. **Cost and maintenance:** The system shall be affordable for the library and maintainable without on-site IT staff. The budget and maintenance expectations need to be defined.
14. **Capacity:** The system shall support the library's current collection of approximately 3,000 books and approximately 800 members.
15. **Future capabilities:** The system's design shall allow for the possible future addition of e-books and inter-library loans.

Open items to validate: The library still needs to decide the overdue policy and clarify reminder timing and method, hold processing, member-account security, measurable cost and maintenance expectations, and whether backup and restore should be a formal requirement. The AI comparison suggested backup and restored but it was not established by the scenario or stakeholder role cards, so it is left for validation rather than included as a confirmed requirement.